ASSIGNMENT-9.3

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Batch: 08

Lab 9: Documentation Generation – Automatic Documentation and Code Comments

Task 1: Basic Docstring Generation

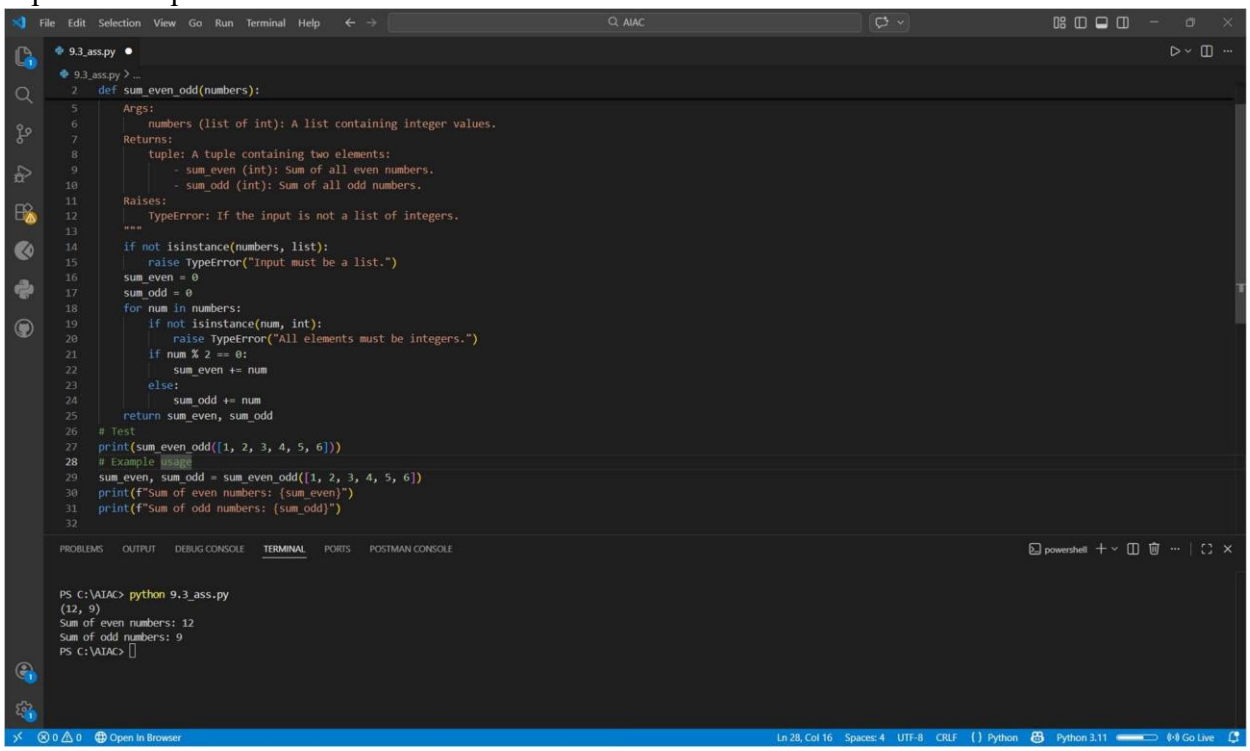
Scenario

You are developing a utility function that processes numerical lists and must be properly documented for future maintenance.

Requirements

- Write a Python function to return the sum of even numbers and sum of odd numbers in a given list
- Manually add a Google Style docstring to the function
- Use an AI-assisted tool (Copilot / Cursor AI) to generate a function-level docstring
- Compare the AI-generated docstring with the manually written docstring
- Analyze clarity, correctness, and completeness

Expected Output



Task 2: Automatic Inline Comments

Scenario

You are developing a student management module that must be easy to understand for new developers.

Requirements

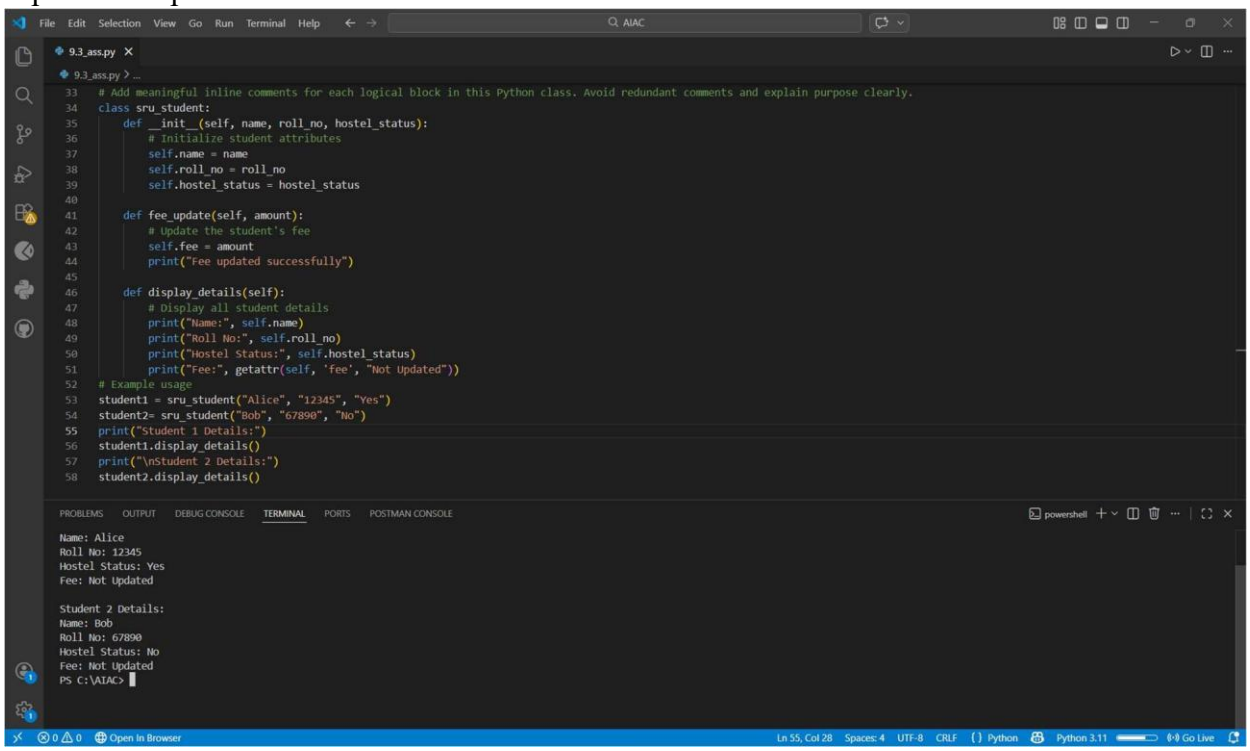
- Write a Python program for an sru_student class with the following:

– Attributes: name, roll_no, hostel_status

– Methods: fee_update() and display_details()

• Manually write inline comments for each line or logical block
• Use an AI-assisted tool to automatically add inline comments
• Compare manual comments with AI-generated comments
• Identify missing, redundant, or incorrect AI comments

Expected Output



Task 3: Module-Level and Function-Level Documentation

Scenario

You are building a small calculator module that will be shared across multiple projects and requires structured documentation.

Requirements

• Write a Python script containing 3–4 functions (e.g., add, subtract, multiply, divide) • Manually write NumPy Style docstrings for each function
• Use AI assistance to generate:
– A module-level docstring
– Individual function-level docstrings
• Compare AI-generated docstrings with manually written ones
• Evaluate documentation structure, accuracy, and readability

Expected Output

```python
#Generate a professional module-level docstring for this calculator module. Use NumPy style. Include description and usage example.
"""
Calculator Module
=================
This module provides basic arithmetic operations such as addition, subtraction, multiplication, and division. It is designed to be simple and easy to use for basic calculations.

"""
def add(a, b):
    """Add two numbers.
    Args:
        a (float): The first number.
        b (float): The second number.
    Returns:
        float: The sum of a and b.
    Example:
        >>> add(2, 3)
        5
    """
    return a + b
def subtract(a, b):
    """Subtract one number from another.
    Args:
        a (float): The number to be subtracted from.
        b (float): The number to subtract.
    Returns:
        float: The difference of a and b.
    Example:
        >>> subtract(5, 2)
        3
    """
    return a - b
def multiply(a, b):
    """Multiply two numbers.
    Args:
        a (float): The first number.
        b (float): The second number.
    Returns:
        float: The product of a and b.
    Example:
        >>> multiply(2, 3)
        6
    """
```

```
PS C:\AIAC> python 9.3_ass.py
PS C:\AIAC> python 9.3_ass.py
5
3
6
3.0
PS C:\AIAC>
```

```python
def multiply(a, b):
    """
        >>> multiply(2, 3)
        6
    """
    return a * b
def divide(a, b):
    """Divide one number by another.
    Args:
        a (float): The numerator.
        b (float): The denominator.
    Returns:
        float: The quotient of a and b.
    Raises:
        ValueError: If b is zero.
    Example:
        >>> divide(6, 2)
        3.0
    """
    if b == 0:
        raise ValueError("Cannot divide by zero.")
    return a / b
print(add(2, 3))
print(subtract(5, 2))
print(multiply(2, 3))
print(divide(6, 2))
```

```
PS C:\AIAC> python 9.3_ass.py
PS C:\AIAC> python 9.3_ass.py
5
3
6
3.0
PS C:\AIAC>
```